

MSMS 407 : Longitudinal Data Analysis

Ananda Biswas

⊕ Restricted Maximum Likelihood Estimation

The method of REML estimation, was introduced by **Patterson** and **Thompson** (1971) as a way of **estimating variance components** in a GLM. The **objection** to the standard maximum likelihood procedure is that it produces **biased estimators of the covariance parameters**. This is well known in the case of the GLM with independent errors,

$$\mathbf{Y} \sim MVN(X\boldsymbol{\beta}, \sigma^2 I). \quad (1)$$

In this case, the maximum likelihood estimator of σ^2 is

$$\widehat{\sigma^2} = \frac{RSS}{mn},$$

where RSS denotes the residual sum of squares, whereas the usual unbiased estimator is

$$\widetilde{\sigma^2} = \frac{RSS}{mn - p},$$

where p is the number of elements of $\boldsymbol{\beta}$. In fact, $\widetilde{\sigma^2}$ is the REML estimator for σ^2 in the model (1).

In the case of the GLM with dependent errors,

$$\mathbf{Y} \sim MVN(X\boldsymbol{\beta}, \sigma^2 V), \quad (2)$$

the REML estimator is defined as a maximum likelihood estimator based on a linearly transformed set of data $\mathbf{Y}^* = A\mathbf{Y}$ such that the distribution of $\mathbf{Y}^*$ does not depend on $\boldsymbol{\beta}$. One way to achieve this is by taking A to be the matrix which converts $\mathbf{Y}$ to OLS residuals (see Appendix A),

$$A = I - X(X'X)^{-1}X'. \quad (3)$$

Then, $\mathbf{Y}^*$ has a **singular** multivariate Gaussian distribution with **mean zero**, whatever the value of $\boldsymbol{\beta}$ (see Appendix B).

To obtain a **non-singular distribution**, we could use only $nm - p$ rows of the matrix, A , defined at (3) (see Appendix C). It turns out that the resulting estimators for σ^2 and V do not depend on which rows we use, nor indeed on the particular choice of A : any full-rank matrix with the property that $\mathbb{E}(\mathbf{Y}^*) = 0$ for all $\boldsymbol{\beta}$ will give the same answer.

Furthermore, from an operational viewpoint we do not need to make the transformation from $\mathbf{Y}$ to $\mathbf{Y}^*$ explicit, for the final estimators do not depend on A . The algebraic details are as follows.

For the theoretical development it is convenient to re-absorb σ^2 into V , and write our model for the response vector $\mathbf{Y}$ as

$$\mathbf{Y} \sim MVN(X\beta, H), \quad (4)$$

where $H \equiv H(\alpha)$, the variance matrix of $\mathbf{Y}$, is characterized by a vector of parameters α , for example, in Uniform correlation model, $\alpha = (\sigma^2, \rho)$.

Our **plan** is to transform $\mathbf{Y}$ to $\mathbf{Z}$ whose distribution does not depend of β and use the likelihood of $\mathbf{Z}$ to estimate the covariance parameters α .

Let A be the matrix defined in (3), and B the $nm \times (nm - p)$ matrix defined by the requirements that $BB' = A$ and $B'B = I$, where I denotes the $(nm - p) \times (nm - p)$ identity matrix. Finally, let

$$\mathbf{Z}_{(nm-p) \times 1} = B'\mathbf{Y}.$$

Note that,

$$\begin{aligned} \mathbb{E}(\mathbf{Z}) &= B'\mathbb{E}(\mathbf{Y}) \\ &= B'X\beta \\ &= \underbrace{B'B}_{I} B'X\beta, \text{ since } B'B = I \\ &= B' \underbrace{BB'}_A X\beta \\ &= B'AX\beta \\ &= B'[I - X(X'X)^{-1}X']X\beta \\ &= B'[X - X(X'X)^{-1}X'X]\beta \\ &= B'[X - X]\beta \\ &= B'\mathbf{0}B \\ &= \mathbf{0}. \end{aligned}$$

$\mathbf{Y} \sim MVN(X\beta, H)$, then $\mathbf{Z} = B'\mathbf{Y} \sim MVN(\mathbf{0}, B'HB)$ and the REML likelihood is

$$L^*(\alpha) = \frac{1}{(2\pi)^{(nm-p)/2} |B'HB|^{1/2}} \exp\left\{-\frac{1}{2} \mathbf{z}^T (B'HB)^{-1} \mathbf{z}\right\}. \quad (5)$$

It can be clearly seen that the resulting estimator $\hat{\alpha}$ will depend on the arbitrary matrix B , hence different choices of B will give different estimators. Also, this representation hides the connection with GLS residuals. So proceeding with the likelihood (5) is not the greatest of ideas. We take a different route.

First note that, for fixed α the maximum likelihood estimator for β is the generalized least-squares estimator,

$$\hat{\beta} = (X'H^{-1}X)^{-1}X'H^{-1}\mathbf{Y} = G\mathbf{Y}, \text{ say,}$$

and

$$\widehat{\beta} \sim MVN(\beta, (X'HX)^{-1}). \quad (6)$$

Secondly recall,

$$\mathbf{Z} = B'\mathbf{Y}.$$

Consider the transformation

$$\mathbf{Y} \mapsto \begin{pmatrix} \mathbf{Z} \\ \widehat{\beta} \end{pmatrix} = \begin{pmatrix} B' \\ G \end{pmatrix} \mathbf{Y} = T\mathbf{Y}, \text{ say.}$$

We write,

$$f_{\mathbf{Z}, \widehat{\beta}}(\mathbf{z}, \widehat{\beta}) = f_{\mathbf{Y}}(\mathbf{y}) \cdot J$$

where J is the Jacobian of the transformation given by

$$J = \frac{1}{\left| \frac{\partial}{\partial \mathbf{Y}} T\mathbf{Y} \right|} = \frac{1}{|T|} = |T|^{-1}.$$

$\mathbf{Z}$ and $\widehat{\beta}$ are **independently distributed** (see Appendix D) and we can write

$$\begin{aligned} f_{\mathbf{Z}}(\mathbf{z}) \cdot f_{\widehat{\beta}}(\widehat{\beta}) &= f_{\mathbf{Y}}(\mathbf{y}) \cdot J \\ \Rightarrow f_{\mathbf{Z}}(\mathbf{z}) &= \frac{f_{\mathbf{Y}}(\mathbf{y})}{f_{\widehat{\beta}}(\widehat{\beta})} \cdot |T|^{-1}. \end{aligned}$$

The Jacobian is free from the parameters β and α (see Appendix E) and thus can be ignored while making inferences.

Finally, we can write

$$f_{\mathbf{Z}}(\mathbf{z}) \propto \frac{f_{\mathbf{Y}}(\mathbf{y})}{f_{\widehat{\beta}}(\widehat{\beta})}. \quad (7)$$

From (4),

$$f_{\mathbf{Y}}(\mathbf{y}) = \frac{1}{(2\pi)^{nm/2} |H|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{y} - X\beta)' H^{-1} (\mathbf{y} - X\beta) \right\}. \quad (8)$$

From (6),

$$f_{\widehat{\beta}}(\widehat{\beta}) = \frac{1}{(2\pi)^{p/2} |X'H^{-1}X|^{-1/2}} \exp \left\{ -\frac{1}{2} (\widehat{\beta} - \beta)' (X'H^{-1}X) (\widehat{\beta} - \beta) \right\}. \quad (9)$$

We state a result,

$$(\mathbf{y} - X\boldsymbol{\beta})' H^{-1} (\mathbf{y} - X\boldsymbol{\beta}) = (\mathbf{y} - X\hat{\boldsymbol{\beta}})' H^{-1} (\mathbf{y} - X\hat{\boldsymbol{\beta}}) + (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})' (X' H^{-1} X) (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}). \quad (10)$$

Using (7), (8), (9) along with (10), we get the PDF of $\mathbf{Z} = B'\mathbf{Y}$ is proportional to

$$\frac{f_{\mathbf{y}}(\mathbf{y})}{f_{\hat{\boldsymbol{\beta}}}(\hat{\boldsymbol{\beta}})} = (2\pi)^{-\frac{1}{2}(nm-p)} |H|^{-1/2} |X' H^{-1} X|^{-1/2} \times \exp \left\{ -\frac{1}{2} (\mathbf{y} - X\hat{\boldsymbol{\beta}})' H^{-1} (\mathbf{y} - X\hat{\boldsymbol{\beta}}) \right\} \quad (11)$$

Note that the right-hand side of (11) is independent of A , and the same result would therefore hold for any $\mathbf{Z}$ such that $\mathbb{E}(\mathbf{Z}) = \mathbf{0}$ and $\text{Cov}(\mathbf{Z}, \hat{\boldsymbol{\beta}}) = \mathbf{0}$.

The practical implication of (11) is that the REML estimator, $\tilde{\boldsymbol{\alpha}}$, maximizes the log-likelihood

$$\ell^*(\boldsymbol{\alpha}) = -\frac{1}{2} \log |H| - \frac{1}{2} \log |X' H^{-1} X| - \frac{1}{2} (\mathbf{y} - X\hat{\boldsymbol{\beta}})' H^{-1} (\mathbf{y} - X\hat{\boldsymbol{\beta}}), \quad (12)$$

whereas the maximum likelihood estimator $\hat{\boldsymbol{\alpha}}$ maximizes

$$\ell(\boldsymbol{\alpha}) = -\frac{1}{2} \log |H| - \frac{1}{2} (\mathbf{y} - X\hat{\boldsymbol{\beta}})' H^{-1} (\mathbf{y} - X\hat{\boldsymbol{\beta}}).$$

It follows from this last result that the algorithm to implement REML estimation for the model (2) incorporates only a simple modification to the maximum likelihood algorithm derived earlier.



Following (2), recall that we consider m units with n measurements per unit and that $\sigma^2 V$ is a block-diagonal matrix with non-zero $n \times n$ blocks $\sigma^2 V_0$ representing the common covariance matrix of the measurements on any one unit.

The original complete log-likelihood is

$$\ell(\boldsymbol{\beta}, \sigma^2, V_0) = -\frac{1}{2} mn \ln(\sigma^2) - \frac{1}{2} m \ln(|V_0|) - \frac{1}{2\sigma^2} (\mathbf{y} - X\boldsymbol{\beta})' V^{-1} (\mathbf{y} - X\boldsymbol{\beta}). \quad (13)$$

In (13), for given V_0 , V will be completely known and we write

$$\hat{\boldsymbol{\beta}}(V_0) = (X' V^{-1} X)^{-1} X' V^{-1} \mathbf{y}. \quad (14)$$

and

$$\text{RSS}(V_0) = \left\{ \mathbf{y} - X\hat{\boldsymbol{\beta}}(V_0) \right\}' V^{-1} \left\{ \mathbf{y} - X\hat{\boldsymbol{\beta}}(V_0) \right\}. \quad (15)$$

Put $H(\boldsymbol{\alpha}) = \sigma^2 V$ in (4) with $\boldsymbol{\alpha} = (\sigma^2, V_0)$. On using (14) and $H(\boldsymbol{\alpha}) = \sigma^2 V$, general (12) becomes special case

$$\begin{aligned}
\ell^*(\sigma^2, V_0) &= -\frac{1}{2} \log |\sigma^2 V| - \frac{1}{2} \log |X'(\sigma^2 V)^{-1} X| - \frac{1}{2} (\mathbf{y} - X\hat{\boldsymbol{\beta}})' (\sigma^2 V)^{-1} (\mathbf{y} - X\hat{\boldsymbol{\beta}}) \\
&= -\frac{1}{2} \log \{(\sigma^2)^{mn} |V|\} - \frac{1}{2} \log \left\{ \left(\frac{1}{\sigma^2} \right)^p |X' V^{-1} X| \right\} - \frac{1}{2\sigma^2} (\mathbf{y} - X\hat{\boldsymbol{\beta}})' V^{-1} (\mathbf{y} - X\hat{\boldsymbol{\beta}}) \\
&= -\frac{1}{2} (mn - p) \log \sigma^2 - \frac{1}{2} m \cdot \log |V_0| - \frac{1}{2} \log |X^T V^{-1} X| - \frac{1}{2\sigma^2} (\mathbf{y} - X\hat{\boldsymbol{\beta}})^T V^{-1} (\mathbf{y} - X\hat{\boldsymbol{\beta}}) \quad (16)
\end{aligned}$$

Then maximizing (16) w.r.t. σ^2 gives the REML estimator for σ^2 as

$$\tilde{\sigma}^2(V_0) = \frac{\text{RSS}(V_0)}{nm - p}, \quad (17)$$

where p is the number of elements of $\boldsymbol{\beta}$. The REML estimator for V_0 maximizes the reduced REML log-likelihood

$$\ell^*(V_0) = -\frac{1}{2} m \{n \log \text{RSS}(V_0) + \log(|V_0|)\} - \frac{1}{2} \log(|X' V^{-1} X|). \quad (18)$$

Finally, maximization of (18) w.r.t. distinct elements of V_0 yields $\tilde{V}_0$. Substitution of the resulting estimator $\tilde{V}_0$ into (14) and (17) gives the REML estimators

$$\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}(\tilde{V}_0) \quad \& \quad \tilde{\sigma}^2 = \tilde{\sigma}^2(\tilde{V}_0).$$

Appendix

A $\mathbf{Y}$ to OLS residuals

In OLS,

$$\hat{\boldsymbol{\beta}} = (X'X)^{-1} X' \mathbf{Y},$$

$$\hat{\mathbf{Y}} = X(X'X)^{-1} X' \mathbf{Y}.$$

Then,

$$\begin{aligned}
\mathbf{e} &= \mathbf{Y} - \hat{\mathbf{Y}} \\
&= \mathbf{Y} - X(X'X)^{-1} X' \mathbf{Y} \\
&= [I - X(X'X)^{-1} X'] \mathbf{Y}.
\end{aligned}$$

B $\mathbf{Y}^*$ has a singular distribution with mean $\mathbf{0}$

$\mathbf{Y}^* = A\mathbf{Y}$, $\mathbf{Y}$ has a multivariate normal distribution, consequently $\mathbf{Y}^*$ also has a multivariate normal distribution.

$$E(\mathbf{Y}^*) = A \cdot E(\mathbf{Y}) = (I - X(X'X)^{-1} X') \cdot X\boldsymbol{\beta} = X\boldsymbol{\beta} - X(X'X)^{-1} X'X\boldsymbol{\beta} = X\boldsymbol{\beta} - X\boldsymbol{\beta} = \mathbf{0}$$

and

$$\text{disp}(\mathbf{Y}^*) = \sigma^2 \cdot AVA'.$$

Then $|\text{disp}(\mathbf{Y}^*)| = \sigma^2 \cdot |V| |A|^2$.

$$\text{Also, } AX = [I - X(X'X)^{-1}X'] X = [X - X(X'X)^{-1}X'X] = [X - X] = \mathbf{O}.$$

$$\therefore A\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} \in \mathcal{N}(A) \Rightarrow \text{Nullity}(A) \neq 0 \Rightarrow \text{rank}(A) < mn \Rightarrow |A| = 0.$$

$$\therefore |\text{disp}(\mathbf{Y}^*)| = 0 \Rightarrow \text{disp}(\mathbf{Y}^*) \text{ has a singular distribution.}$$

C Error Contrast

A linear combination $\mathbf{a}'\mathbf{Y}$ of the response variable $\mathbf{Y}$ is called an **error contrast** if $\mathbb{E}(\mathbf{a}'\mathbf{Y}) = 0$. For a linear model $\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$, the condition reduces to $\mathbf{a}'X = \mathbf{0}$.

It readily follows that each row of A as in (3) is an error contrast as $AX = \mathbf{O}$. Recall the fact that, given n error contrasts, at max $n - p$ can be linearly independent, p being the rank of the related design matrix X . $\mathbf{Y}^* = A\mathbf{Y}$ is then a vector of error contrasts as each row of A produces an error contrast. Thus to have a nonsingular distribution of $\mathbf{Y}^*$ we must take $mn - p$ rows of A , of course the $mn - p$ rows must be linearly independent.

D $\mathbf{Z}$ and $\hat{\boldsymbol{\beta}}$ are independent

$$\begin{aligned} \text{Cov}(\mathbf{Z}, \hat{\boldsymbol{\beta}}) &= \mathbb{E} \left\{ \mathbf{Z} (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})' \right\} \\ &= \mathbb{E} \left\{ B'\mathbf{Y} (\mathbf{Y}'G' - \boldsymbol{\beta}') \right\} \\ &= B'\mathbb{E} (\mathbf{Y}\mathbf{Y}') G' - B'\mathbb{E} (\mathbf{Y}) \boldsymbol{\beta}' \\ &= B' \{ \text{Var}(\mathbf{Y}) + \mathbb{E}(\mathbf{Y})\mathbb{E}(\mathbf{Y})' \} G' - B'\mathbb{E}(\mathbf{Y})\boldsymbol{\beta}'. \end{aligned}$$

Now, substituting $\mathbb{E}(\mathbf{Y}) = X\boldsymbol{\beta}$ into this last expression gives

$$\text{Cov}(\mathbf{Z}, \hat{\boldsymbol{\beta}}) = B' (H + X\boldsymbol{\beta}\boldsymbol{\beta}'X') G' - B'X\boldsymbol{\beta}\boldsymbol{\beta}'.$$

Also,

$$\begin{aligned} X'G' &= X'H^{-1}X (X'H^{-1}X)^{-1} \\ &= I, \end{aligned}$$

and

$$\begin{aligned} B'HG' &= B'HH^{-1}X (X'H^{-1}X)^{-1} \\ &= B'X (X'H^{-1}X)^{-1} \\ &= 0, \end{aligned}$$

because $B'X = B'AX = \mathbf{O}$ as in the proof that $\mathbb{E}(\mathbf{Z}) = \mathbf{0}$. It follows that

$$\begin{aligned}\text{Cov}(\mathbf{Z}, \widehat{\boldsymbol{\beta}}) &= B'X\boldsymbol{\beta}\boldsymbol{\beta}' - B'X\boldsymbol{\beta}\boldsymbol{\beta}' \\ &= \mathbf{O}.\end{aligned}$$

Finally, in the multivariate Gaussian setting, zero covariance is equivalent to independence.

E $|T|$

$$\begin{aligned}|T| &= |TT'|^{1/2} \\ &= \left[\begin{pmatrix} B' \\ G \end{pmatrix} (B \ G') \right]^{1/2} \\ &= \begin{vmatrix} B'B & B'G' \\ GB & GG' \end{vmatrix}^{1/2} \\ &= \begin{vmatrix} I & B'G' \\ GB & GG' \end{vmatrix}^{1/2} \quad \text{as } B'B = I \\ &= |I|^{1/2} \cdot |GG' - GB I^{-1} B'G'|^{1/2} \\ &= |GG' - GBB'G'|^{1/2} \\ &= |GG' - GAG'|^{1/2} \quad \text{as } BB' = A \\ &= |GG' - G[I - X(X'X)^{-1}X']G'|^{1/2} \\ &= |GG' - [GG' - GX(X'X)^{-1}X'G']|^{1/2} \\ &= |GG' - GG' + GX(X'X)^{-1}X'G'|^{1/2} \\ &= |GX(X'X)^{-1}X'G'|^{1/2} \\ &= |(X'X)^{-1}|^{1/2} \quad \text{as } GX = I \\ &= |X'X|^{-1/2}.\end{aligned}$$